Jingsi Zhou

(210) 862-0768 | jingsi.zhou66@gmail.com | linkedin.com/in/jingsi-zhou | tartlet.github.io

PROFESSIONAL EXPERIENCE

York Space Systems **May 2026–Present**

Senior Software Engineer

- Design and evolve software-in-the-loop (SITL) and processor-in-the-loop (PITL) infrastructure and an internal message-parsing library with a focus on maintainability, backward compatibility, and stable user workflows.
- Develop and review integration and test improvements, including automated safety checks before PITL execution to help prevent overheating from high-current conditions.
- Lead technical roadmaps for two spacecraft missions, tracking milestones, releases, simulation deliverables, test events, component gaps, and cross-team priorities.
- Serve as the technical point of contact for SITL/PITL setup and staging/sandbox-environment deployments.
- Diagnose and resolve integrated flatsat hardware and network issues across data logging, flight-computer connectivity, time synchronization, and component communications.
- Build and integrate a telemetry data-injection tool into the SITL stack, enabling Mission Operations to simulate in-orbit anomalies and malfunctions for training.

York Space Systems **Oct 2023–May 2026**

Software Engineer

- Implemented, integrated, and tested simulated spacecraft components in full SITL environments, replacing physical hardware with software models for end-to-end validation.
- Helped automate PITL setup across vehicles and flatsats, replacing 10+ minutes of manual JSON editing and transfer to flight computers with a flag-driven Bash workflow adopted by multiple teams and missions.
- Troubleshot GitLab CI/CD pipelines supporting L2–L4 test stages (integration, system, and acceptance testing), improving reliability of automated validation.
- Applied a "test like you fly" approach to make SITL and PITL environments more representative of flight and production operations; used SITL testing to identify flight-software defects before deployment.
- Maintained simulation components, test environments, and documentation through flight-software updates, preventing configuration and version drift.
- Supported complex demonstrations, mission rehearsals, and test campaigns through environment preparation, execution coordination, troubleshooting, and follow-through.

Amazon Project Kuiper **May 2022–Aug 2022**

Embedded Software Development Intern

- Designed, developed, and documented a solar-array firmware test script.
- Diagnosed and corrected defects in Rust-based solar-array firmware.
- Collaborated with mechanical and electrical engineering teams on hardware issues and testing within an Agile cadence.

SELECTED PROJECTS

StonkR – *Full-Stack Developer* **Jan 2023–Apr 2023**

Built and deployed a stock-tracking application with React, Python, PostgreSQL, and GCP; documented and tested an internal API with Postman.

EDUCATION

The University of Texas at Austin **May 2023**

B.S. Mechanical Engineering · Minor in Computer Science

Overall GPA: 3.89/4.00 · Major GPA: 3.92/4.00

TECHNICAL SKILLS

Languages: C, C++, Python, JavaScript, Swift, MATLAB, Rust

Infrastructure & Networking: Docker, WSL2, Linux, Ansible, GitLab CI/CD, iptables routing, network bridges, subnets

Data & Applications: MySQL, PostgreSQL, SQLAlchemy, Flask, React, Node.js, GraphQL, Firebase, GCP

Tools & Methods: Git, Jira, Confluence, Postman, Agile/Scrum, API documentation, SolidWorks, Fusion 360